

Axial Flow compressor
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Given Data,

Mass flow rate @ inlet = 131.2447 kg/s

Rotational speed 'N' (rpm) = 373.56 m/s

Inlet stag. temp (T_{01}) = 389.86 K

Inlet stag. pressure (P_{01}) = 73.66 kPa

Static pressure $P_1 = 60.05 \text{ kPa}$

Flow coefficient, $\phi = \frac{C_a}{U} = 0.4$

Polytropic efficiency (η_p) = 0.8782

* Basic Design procedure:

1. Inlet static conditions & Annulus sizing.

2. Stage by stage Design

3. Exit Annulus sizing & Air angle Variation

Plot Vel Δ les, De Haller number, Deflection in every stage.

Plot Tables in step 3 as graph.

No IGV

Inlet swirl

$\alpha_1 = 0$

Hub-tip ratio, $\frac{r_h}{r_t} = 0.27$

Mach number = 0.549
@ inlet

Total PR = 4.1

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Step ①:

Inlet static conditions:

$$T_{01} = T_1 \left(1 + \frac{\gamma-1}{2} \right) M^2 \Rightarrow T_1 = \frac{T_{01}}{\left(1 + \frac{\gamma-1}{2} \right) M^2}$$

$$\rho = \rho_1 = \frac{P_1}{R \times T_1}$$

$$\dot{m} = \rho_1 C_a A_1 = \rho_1 C_a \pi (r_t^2 - r_r^2) \Rightarrow \rho_1 C_a \pi r_t^2 (1 - \phi^2)$$

$$r_t^2 = \frac{\dot{m}}{\rho_1 C_a \pi (1 - \phi^2)}$$

$$\phi = \frac{r_r}{r_t} \Rightarrow \dots$$

$$\phi = \frac{C_a}{U} \Rightarrow \dots$$

$$r_r = \phi \times r_t$$

$$r_{\text{mean}} = r_m = \frac{r_r + r_t}{2}$$

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Step ②: Stage by stage design

i, Stage 1:

Assume, $\Delta T_0 \approx 20K$

$$\lambda_1 = 0.98$$

$$\alpha_1 = 0 \Rightarrow Cw_1 = 0$$

$$\Delta Cw = Cw_2 - Cw_1 = \frac{Cp \Delta T_0}{\lambda U} \Rightarrow Cw_2 = \Delta Cw$$

$$\text{Then, } \tan \beta_1 = \frac{U}{Ca} \rightarrow \beta_1$$

$$\text{Deflection: } \beta_1 - \beta_2$$

$$\tan \beta_2 = \frac{U - Cw_2}{Ca} \rightarrow \beta_2$$

$$\text{De Haller Number: } \frac{V_2}{V_1} = \frac{\cos \beta_1}{\cos \beta_2}$$

$$\tan \alpha_2 = \frac{Cw_2}{Ca} \rightarrow \alpha_2$$

Pressure Ratio,

$$PR = \left(\frac{P_{03}}{P_{01}} \right)_1 = \left[1 + \eta_p \frac{\Delta T_0}{T_{01}} \right]^{\frac{\gamma}{\gamma-1}}$$

Then,

$$(P_{03})_1 = PR \times (P_{01})_1$$

$$(T_{03})_1 = \Delta T_0 + T_{01}$$

Degree of Reaction,

$$\Lambda = 1 - \left[\frac{Cw_2 + Cw_1}{2U} \right]$$

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ii, Stage 2:

Assume, $\Delta T_{02} = 25 \text{ K}$

$$\lambda_2 = 0.93$$

$$\Delta T_{02} = \frac{\lambda U C_a (\tan \beta_1 - \tan \beta_2)}{C_p} \Rightarrow \tan \beta_1 - \tan \beta_2 = ? \quad (1)$$

$$\text{We know } \Lambda = \frac{C_a}{2U} (\tan \beta_1 + \tan \beta_2) \Rightarrow \tan \beta_1 + \tan \beta_2 = ? \quad (2)$$

From (1) & (2); β_1, β_2

$$\text{Also, } \tan \alpha_1 + \tan \beta_1 = \frac{U}{C_a} \Rightarrow \tan \alpha_1 = ? \rightarrow \alpha_1$$

$$\tan \alpha_2 - \tan \alpha_1 = \tan \beta_2 - \tan \beta_1 \Rightarrow \tan \alpha_2 = ? \rightarrow \alpha_2$$

Whirl vel., $C_{w1} = C_a \tan \alpha_1$, $C_{w2} = C_a \tan \alpha_2$

Pressure Ratio,

$$PR_2 = \left(\frac{P_{03}}{P_{01}} \right)_2 = \left[1 + \eta_p \frac{\Delta T_{02}}{T_{02}} \right]^{\frac{\gamma}{\gamma-1}}$$

$$(P_{03})_2 = PR_2 \times (P_{01})_2 = \text{---}$$

$$(T_{03})_2 = (T_{01})_2 + (\Delta T_0)_2 = \text{---}$$

Reflection: $\beta_1 - \beta_2$

De Haller Number $\Rightarrow \frac{U_2}{U_1} = \frac{\cos \beta_1}{\cos \beta_2}$

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iii Stage (3):

Assume, $\Delta T_{03} = 22.8 \text{ K}$

$$\lambda = 0.88$$

$$\Lambda = 0.5$$

Now, $\tan \beta_1 - \tan \beta_2 = \frac{\Delta T_{03} C_p}{\lambda U C_a} \quad | \quad \beta_1, \beta_2$

$$\tan \beta_1 + \tan \beta_2 = \Lambda \frac{2U}{C_a} \quad | \quad \left(\frac{\gamma}{\gamma-1} \right)$$

$$(PR)_3 = \left(\frac{P_{03}}{P_{01}} \right)_3 = \left[1 + \frac{\eta_p \Delta T_{02}}{T_{01}} \right] \left(\frac{\gamma}{\gamma-1} \right)$$

De Haller

$$\frac{U_2}{U_1} = \frac{\cos \beta_1}{\cos \beta_2}$$

Deflection:

$$\beta_1 - \beta_2$$

$$(P_{03})_3 = PR_3 \times (P_{01})_3 =$$

$$(T_{03})_3 = (T_{01})_3 + (\Delta T_{03}) =$$

Since, $\Lambda = 50\%$

By Symmetry of Velocity Ales $\alpha_1 = \beta_2, \alpha_2 = \beta_1$

$$C_{w1} = C_a \tan \alpha_1 \quad C_{w2} = C_a \tan \alpha_2$$

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iv, Stages 4, 5, 6:

• Stage 4:

Assume, $\lambda = 0.83$

$\Lambda = 50\%$

$\Delta T_{OS} = 22.2\text{K}$

$$\left. \begin{aligned} \tan \beta_1 - \tan \beta_2 &= \frac{\Delta T_{OS} C_p}{\lambda U C_a} \\ \tan \beta_1 + \tan \beta_2 &= \Lambda \frac{2U}{C_a} \end{aligned} \right\} \beta_1, \beta_2$$

$$(PR)_x = \left[1 + \eta_p \cdot \frac{\Delta T_0}{T_{01}} \right]^{\left(\frac{\gamma}{\gamma-1} \right)}$$

$$(P_{03}) = (P_{01}) \times PR$$

$$T_{03} = T_{01} + \Delta T_0$$

Since $\Lambda = 0.5$

$$\beta_1 = \alpha_2, \alpha_1 = \beta_2$$

$$C_{w1} = C_a \tan \alpha_1$$

$$C_{w2} = C_a \tan \alpha_2$$

$$\text{De Haller: } \frac{V_2}{V_1} = \frac{C_{w1} \beta_1}{C_{w2} \beta_2}$$

Deflection: $\beta_2 - \beta_1$

• Use P_{03}, T_{03} as P_{01}, T_{01} in next stage

Iterate and fill Table

	4	5	6
P_{01}			
T_{01}			
PR			
P_{03}			
T_{03}			

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✓ Stage 7: REVERSE ENGINEER

Given Data, Total PR = 4.1

$$= \prod_1^7 (PR)_i$$

$$(PR)_7 = \frac{\text{Total PR}}{\prod_1^6 (PR)_i}$$

Then, $(P_{07})_7 = (PR)_7 \times (P_{01})_7 = \dots$

$$(PR)_7 = \left(1 + \eta_p \frac{\Delta T_0}{T_{07}} \right)^{\frac{\gamma}{\gamma-1}} \Rightarrow \Delta T_0 = \dots$$

$$\text{Then, } \tan \beta_1 - \tan \beta_2 = \frac{\Delta T_{07} C_p}{\lambda l C_a}$$

$$\tan \beta_1 + \tan \beta_2 = \frac{\lambda 2U}{C_a}$$

De Haller

$$\beta_1, \beta_2 \Rightarrow \frac{V_2}{V_1} = \frac{\cot \beta_1}{\cot \beta_2}$$

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Step (3) Exit Annulus Sizing; Air Angle Variation

$$C_3^2 = C_a^2 + C_{w2}^2$$

$$T_3 = T_{03} - \frac{C_3^2}{2C_p}$$

$$P_3 = P_{03} \left[\frac{T_3}{T_{03}} \right]^{\frac{\gamma}{\gamma-1}}$$

$$\rho_3 = \frac{P_3}{R \times T_3}$$

$$A = \frac{\dot{m}}{\rho_3 \times C_{a3}}$$

$$h = \frac{A}{2\pi r_m}$$

$$r_{root} = r_m - \frac{h}{2}$$

$$r_{tip} = r_m + \frac{h}{2} \quad \text{Gill rate}$$

Stage	T_3	P_3	ρ_3	A	r_{root}	r_{tip}
1						
2						
...						
3						

(Only stage 1, $\alpha_1 = 0^\circ$)

$$\tan \beta_1 = \frac{U}{C_a} - \tan \alpha_1^0$$

$$U = r\omega \rightarrow C_{men, U = tip}$$

	β_1	β_2	α_1	α_2
Rotor				
mean				
Tip				

$$\beta_1 = \frac{U}{C_a}$$

$$\beta_2 = \frac{U - C_{w2}}{C_a}$$

$$\beta_3 = \frac{C_{w2}}{C_a}$$

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